Intel Xeon 6 P-Core chips

Sights set to outpace AMD and boost AI data centre business

—By Jayesh Shinde

Intel hasn't had it easy of late. Between internal tumult and aggressive competition from both AMD and NVIDIA, the semiconductor giant's been navigating choppy waters. But with the launch of its new Xeon 6 processors – featuring performance-focused P-cores – Intel is trying to make a statement. It's not just around to survive, but wants to reclaim the throne as the go-to name in data center innovation.

While many pegged the once-dominant CPU maker as lagging behind, the Xeon 6 P-core rollout makes it clear that Intel still wields some serious engineering chops – and a willingness to move the ball forward even under pressure. It's trying to do that in the only way it knows best – championing the cause of x86 in the data centre by addressing its evolving demands for robust AI workloads, serious networking muscle, and more eco-friendly computing.

Intel Xeon 6: E-core vs P-core

In June 2024, Intel unveiled Xeon 6 E-core chips at Computex 2024. Now come the P-cores. One might say Intel's filling out the product line to ensure no matter the workload – heavy AI or general-purpose computing – there's a Xeon solution for enterprise and business customers of all shapes and sizes. That's a broader stroke to remain top-of-mind in data centre designs.

While E-core chips cater to specialised tasks, the P-cores are pitched as the all-around workhorse. If you're a cloud provider, you might want the best of both worlds: specialised E-cores for certain tasks, and broad-range P-cores for others.

We can't ignore the fact that Intel's grip on the data centre scene has slipped a bit. AMD's EPYC chips have carved out a chunk of market share, and NVIDIA is expanding into CPU territory, too. The broadening of Intel's Xeon lineup is a direct message that Intel's technology stack still has the depth to meet all data centre needs. It's Intel's reassertion of a leadership that some thought was disappearing fast.

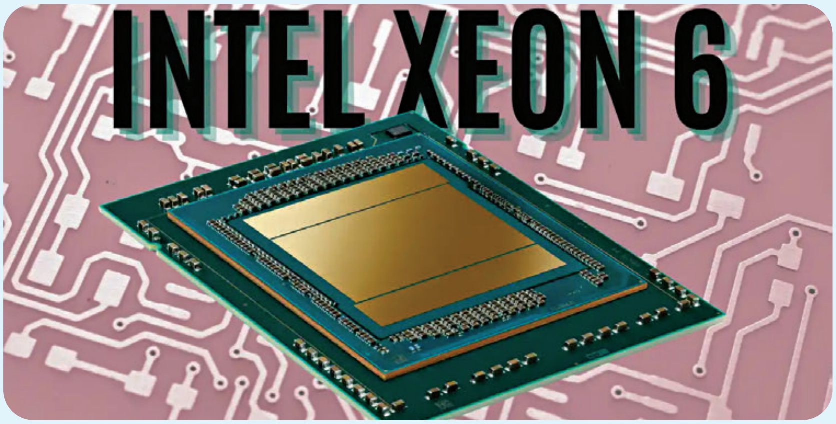
Tuned for AI performance

The Intel Xeon 6700/6500 series processor with P-cores is touting up to a 2x AI processing speed bump versus prior generations, with a typical 1.4x performance lift in other core tasks. And it's showcasing benchmark scores to emphasise this point, which in the real world translates into serious headroom for data centres juggling everything from generative AI to standard enterprise apps. In an era where microseconds can decide competitive advantage, that edge is important. According to Intel, the new Xeons aim to handle both "vanilla" tasks (like basic virtualisation) and advanced AI training or inference sessions. If you're a big retailer looking to parse real-time stock data or run complex recommendations, this sort of horsepower can slash decision times dramatically.

Intel also claims the new Xeon P-core chips deliver a level of consolidation previously only reserved for specialized hardware, potentially saving both rack space and power bills.

Optimized for 5G and Edge

Intel's marketing mentions "built-in accelerators," including Intel vRAN Boost, specifically targeting radio access network (RAN) performance. That's crucial in a 5G era, where networks and edge deployments require minimal latency. A 2.4x capacity bump in RAN tasks is no small feat – particularly for telcos trying to handle ever-rising data loads. Edge computing is, arguably, the next big leap in reducing latency and improving real-time analytics. Everyone from LLM and foundational AI model makers to big tech enterprises believes that more and more intelligence and data processing is happening not inside remote data centres but devices at the edge, from smartphones and laptops to small local servers inside offices. Intel's design choices on the Xeon 6 P-core chips – like integrated accelerators for media processing or network security – are meant to let data get processed closer to the source. For use cases such as autonomous cars or smart factories, that "local intelligence" can be make-or-break.



CONTINUE READING ON DIGIT AI ZED